



Portable Film Digitization System

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Resurgence of Analog Film Photography

Cultural Impact

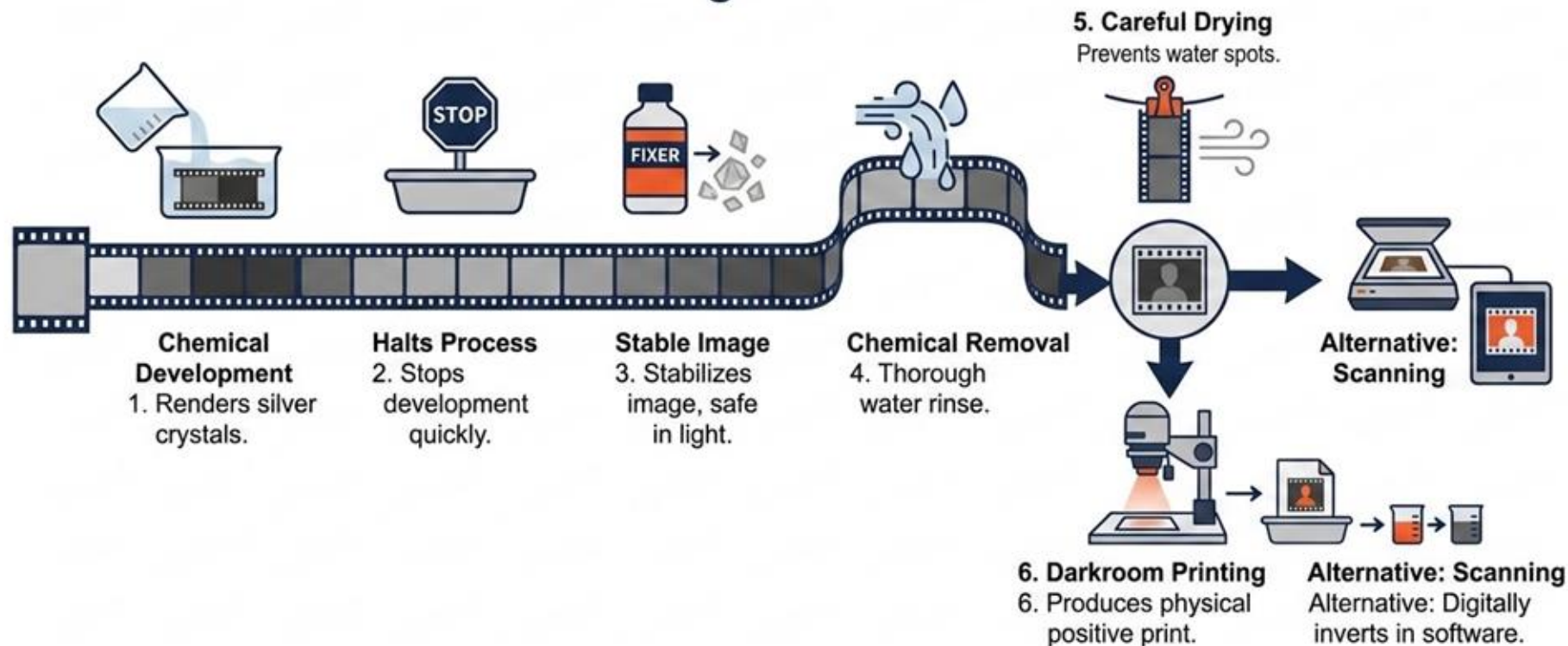
Analog film is experiencing a resurgence, praised for its unique color palette and nostalgic feel—qualities that digital photography struggles to replicate

Barriers to Access

Yet, the complexity of developing and digitizing film is a major hurdle, limiting its accessibility and preventing many from enjoying its full potential.

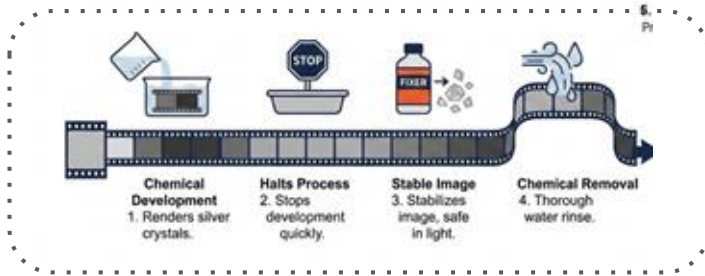


Film Negative Process



Existing Solution – Instax Film

Preprocessing



Post-Processing (Digitalization)



Professional Scanner

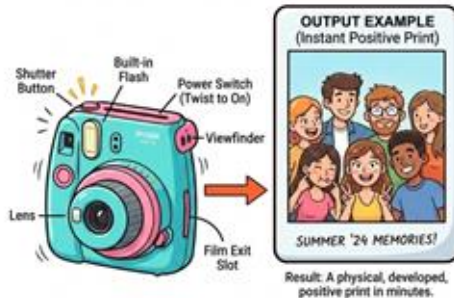
Remain complex

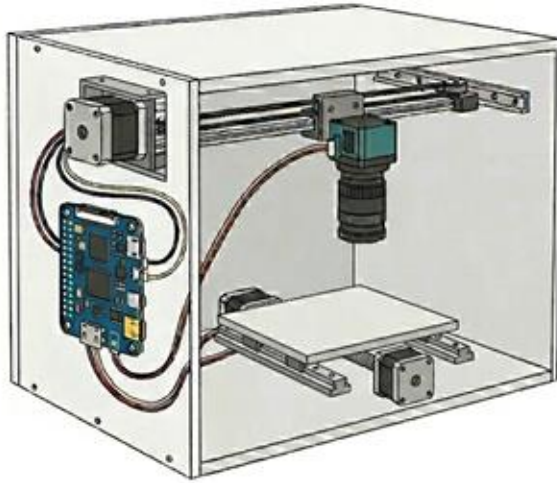
Core Problem

Existing post processing pipelines remain complex, expensive and time consuming!



Film! Rephotography set





Prototype Design

Our Solution: A Portable Digitizer

One-Click Automated Scanning

Our device moves a camera over the surface of analog film to take several high-resolution frame captures.

We then use a computer vision pipeline to stitch these frames together into a final, high-quality digital output.

Functional Requirements

check_circle Stitch together multiple macro-frame captures with minimal alignment error

check_circle Remain portable and accessible as a “plug and play” device

check_circle Complete a full scan in a time frame of 3-4 minutes

Features

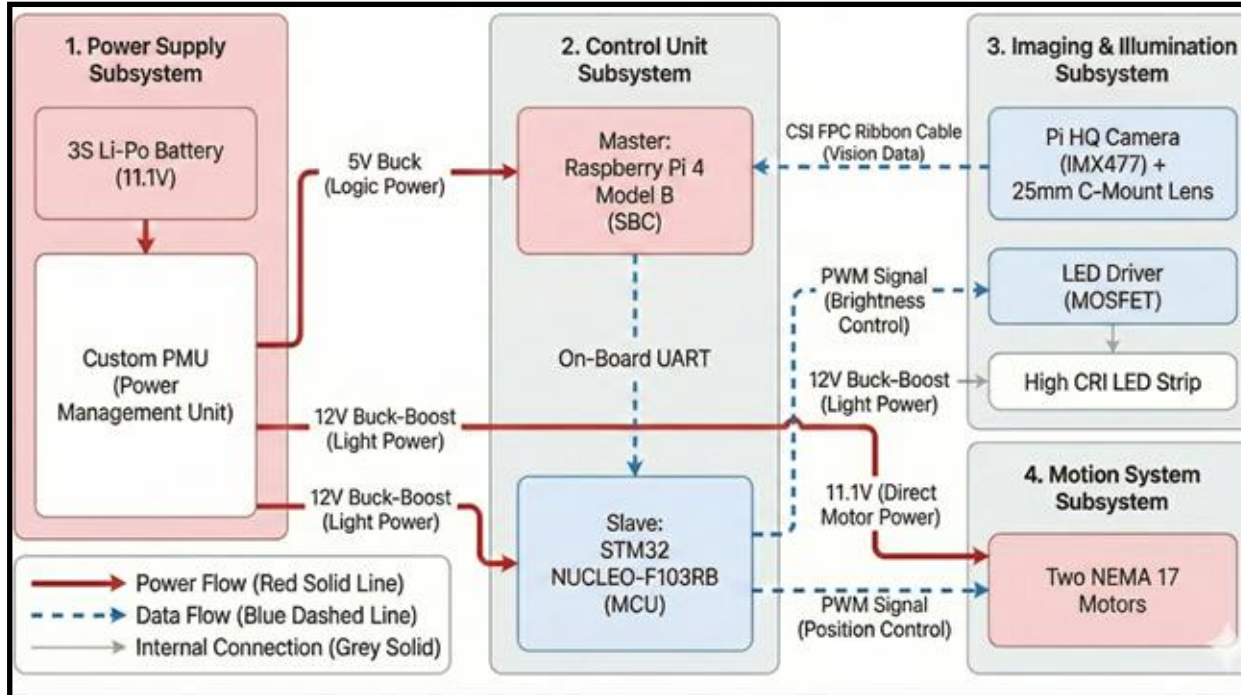
star Support for multiple film formats

star Dual-mode lighting

star High-resolution captures

star Cloud processing pipeline

star Artificial Intelligence Integration



System Block Diagram

Power System Architecture

Core Power & Processing:

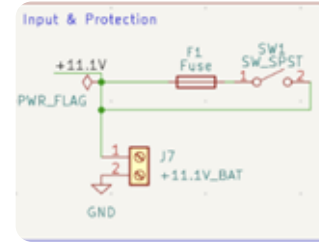
- Main Controller Supply
- Battery & Portability
- Voltage Regulation

Sub-System Power Distribution:

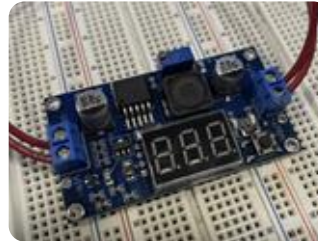
- Circuit Input Power (11V)
- Motor Logic Power (5V)
- Lighting Power (5V)



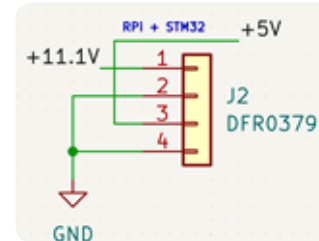
Battery



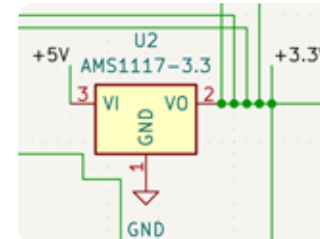
Input Schematics



Converter 1



Converter 1 Schematics



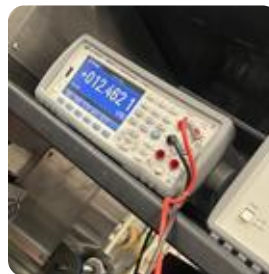
Converter 2 Schematics

Power Regulation

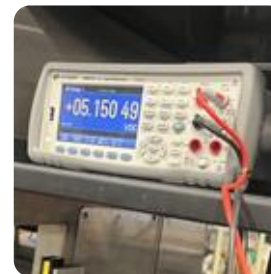
Make sure the battery outputs the source voltage the circuit expects

Verification

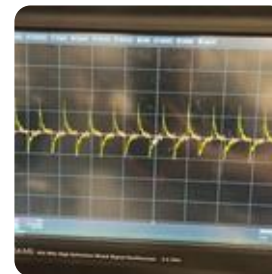
- Apply battery input
- Measure outputs with multimeter
- Run motors at peak load
- Measure logic rail ripple with oscilloscope



Battery Voltage



Logic Voltage



Voltage Ripple

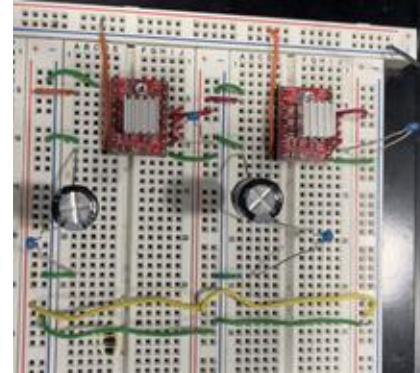
Error Requirement

- Battery: $12\text{ V} \pm 1\text{ V}$ (under 100 mA–2 A load)
- Logic: $\leq 5\text{ V} \pm 0.25\text{ V}$ (under 100 mA–2 A load)
- Logic rail: $\leq 50\text{ mV}$ ripple (peak-to-peak) during operation

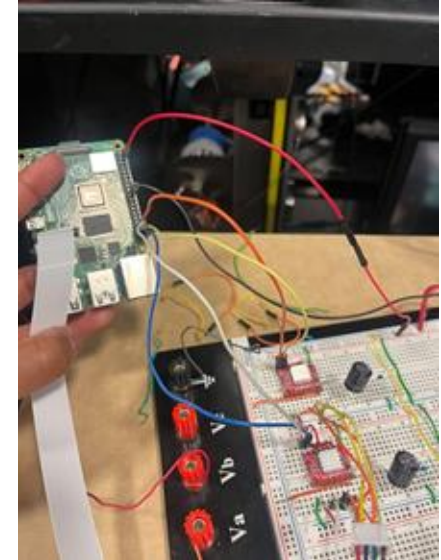
Measured Values

We used lab equipment to verify our circuit on a breadboard and PCB to ensure that the experimental values matched what we expected.

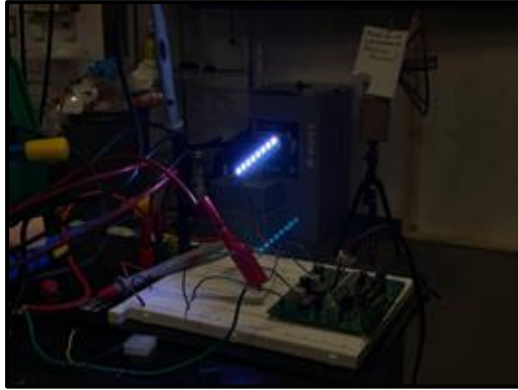
Signal	Value
Battery Input	12.462 V
Regulated Logic	5.150 V
V _{mot}	11.146 V
I _{motor}	0.62 A



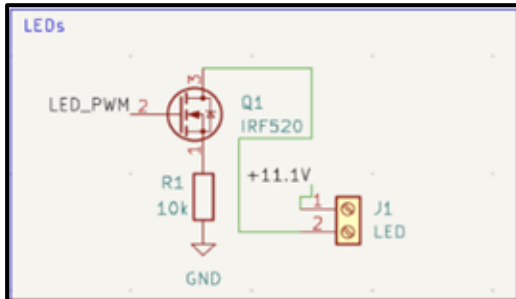
Circuit Prototyping



Raspberry Pi 4 with Circuit



Smart LED Strip



LED Schematics



LED Pad

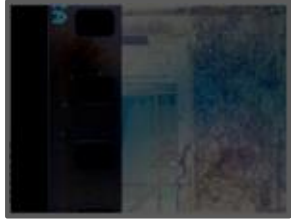
Captures & Smart Lighting

High-Fidelity Imaging Setup

- Core Sensor
- Optics
- Macro Adaptation

Dual-Mode Illumination

- Transmissive Media (Negatives & Slides)
- Reflective Media (Instax Prints)
- Automated Control



10% duty cycle



50% duty cycle



90% duty cycle

Error Requirement

- Lens to film distance: $8.5 \text{ cm} \pm 1 \text{ mm}$
- PWM control: Adjust brightness via PWM without introducing artifacts



Distance between lens and film

Focal Distance Control

Maintain appropriate lens to film distance for image clarity

Verification

- Mount camera + lens
- Capture test image at target distance
- Check image sharpness (visual)

Backlight PWM Control

Ensure no image artifacts

Verification

- Capture images at **10%, 50%, 90% duty cycles**
- Inspect frames for **artifacts/distortion**



Scanning System Components

Precise 2D Motion

We have stepper motors in 2 planes: X and Y. We need to coordinate the movement of both motors to get accurate captures of specific coordinates within the film.

Control Interface:

Raspberry Pi → STM32 → Motor Driver → Motor Coils

- We use motor drivers to control motor speed and direction via digital logic
- Raspberry Pi issues high level commands to MCU over serial
- MCU GPIO pins output STEP and DIR signals
- STEP frequency controls motor speed, DIR controls direction

Motor Drift

Even small movement errors result in drastic image capture errors.

Requirement and Verification

We used the microstepping functionality of our motor drivers to achieve better translational resolution.

- Took a full raster scan and commanded both motors return to their origin positions
- Used firmware and current tuning to end in a state where error was too small to physically measure (< 0.1 mm)
- Measured alignment error of stitched image to be < 5 px

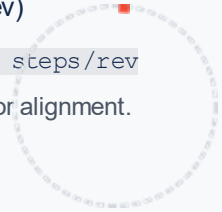
Microstepping Resolution

Full Step: 1.8° (200 steps/rev)

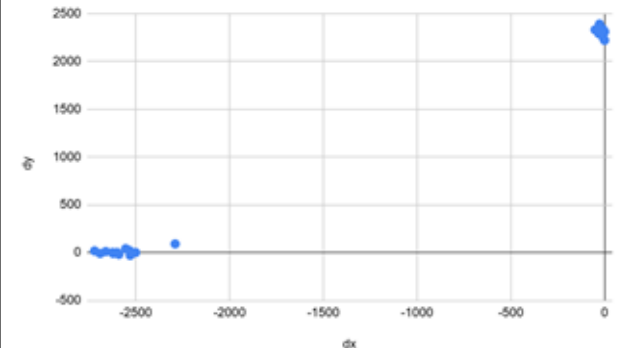
Microsteps: 1/16 (3,200 steps/rev)

$\text{Resolution} = \text{Pitch} / 3,200 \text{ steps/rev}$

Achieves sub-millimeter precision for alignment.



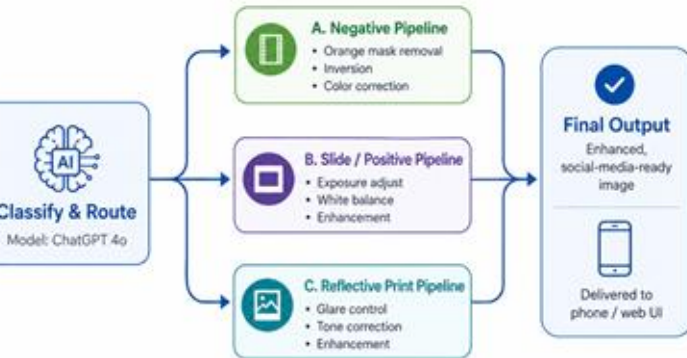
Stitch Offsets (dx vs dy)



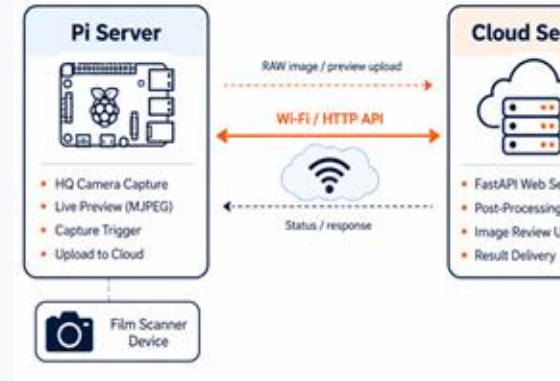
Routing Logic

Post-Processing Overview

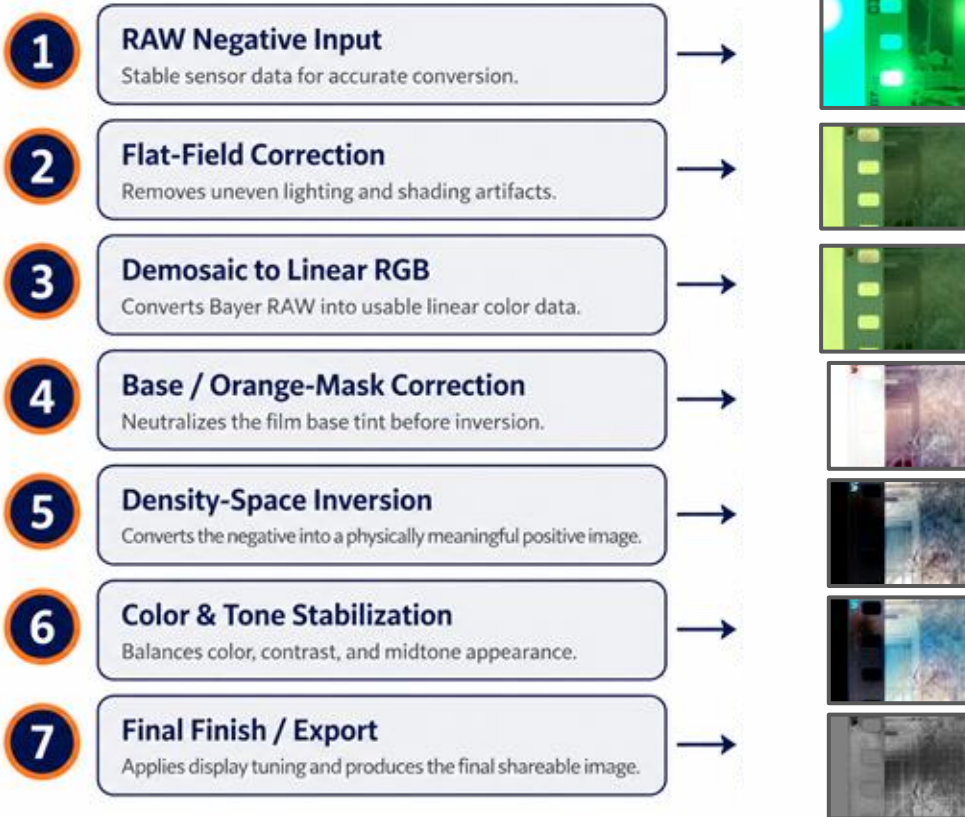
Input → Classify → Route to Pipeline



Server Structure



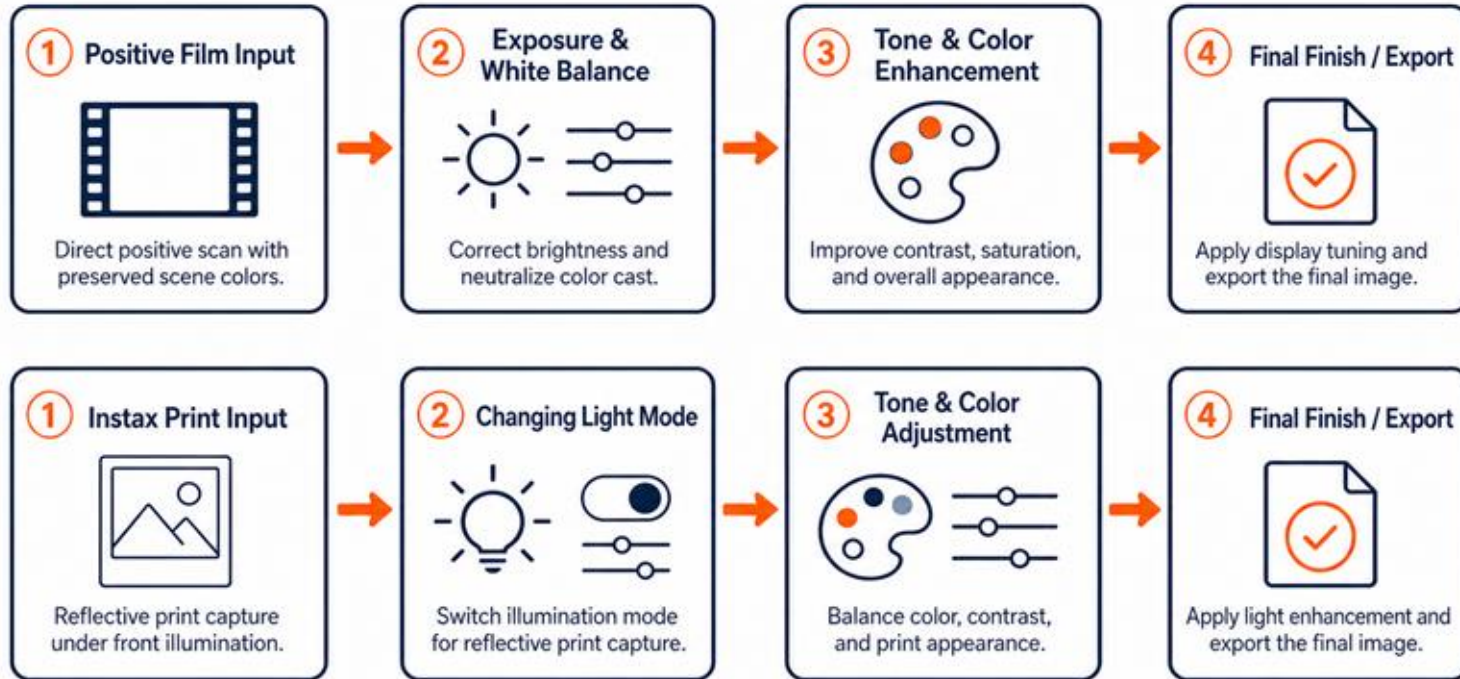
Post-Processing Pipeline: Negative



A simplified pipeline that follows industry standard professional software Negative Lab Pro



Post-Processing Pipeline: Positive+Instant



Coordination and Digital Representation

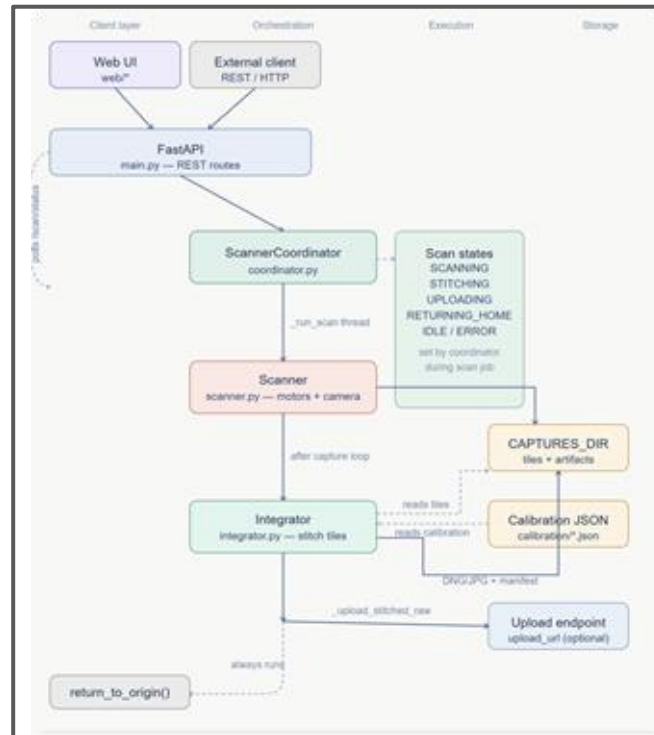
Tying all our subsystems together and coordinating hardware through software was one of the greatest challenges we faced.

State Tracking & Robustness

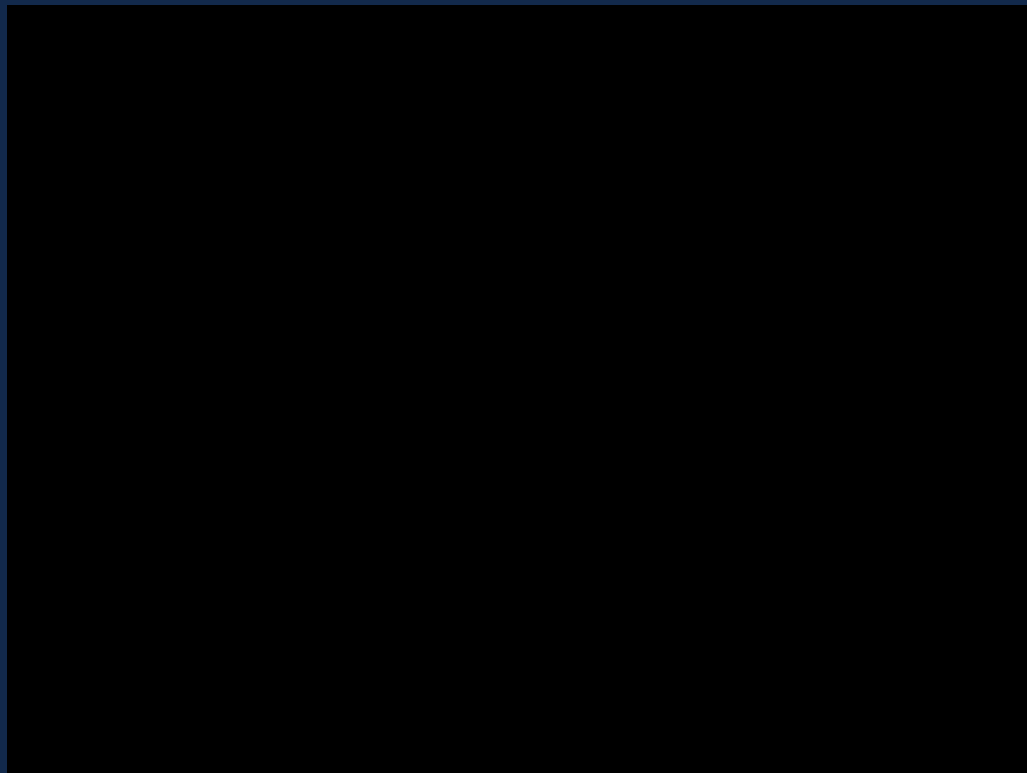
Our system has many failure points:

- Stuck motors
- Power outage – battery discharges
- UART disconnect
- Network errors

By defining integration points and clear “owners” for each piece of logic, we could effectively debug and develop in parallel.



Demo



Failures and Challenges

- Soldering and verifying functionality on PCB
- PCB layout and component selection
- Time management for development and part orders

Successes

- Testing changes in isolation
- Defining integration points before working on implementation
- Making tradeoffs and implementation decisions to support bottom line

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Revisiting Design

Although we were able to build a functional prototype, we identified key areas for improvement to make the device more robust and convenient to use.

Simplifying Circuit Design

Separate logic circuits instead of having a single large one with varying physical requirements

Component Selection

Use well-documented parts with practical footprints.

Calibration Hardware

Add motor limit switches for fully automatic calibration.

Time Management

Invest more time in early design and requirements.

Safety and Battery Management

- Immediate kill switch and insulated wires
- Battery management system (regulator/fuse)
- Sealed enclosure for mechanical pinch points

Standards & Compliance

- Wi-Fi networking compliance
- JPEG standards for data transfer

Misuse Statement

- Warnings against scanning of illegal materials, scanning without permission, or attempting to circumvent copyright rules

Library Attributions

This project leverages several libraries for computer vision, camera control, and more

- OpenCV
- Libcamera
- FastAPI
- ReactJS

Portability, UX, and Speed

Physical Design & Portability

Refine for a smaller footprint and utilize lighter, more durable materials.

Automated Calibration

Incorporate dedicated sensors to streamline and automate the calibration process.

Integrated UI / UX

Add a touchscreen interface to remove the dependency on external phones or laptops.

Software & Speed Optimization

Optimize software algorithms to significantly reduce overall processing time.

Advanced Vision Processing

Leverage specialized hardware accelerators or GPUs for high-speed vision processing tasks.

Questions



Thanks for listening!